

Hayden Farrell

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EDUCATION

University of Nottingham

BSc Computer Science with a Year in Industry (Predicted First-Class Honours avg. 75%)

Nottingham, UK

Aug. 2023 – May 2027

Aquinas College

Mathematics (A), Computer Science (A), Further Mathematics (A)

Manchester, UK

Aug. 2021 – May 2023

EXPERIENCE

Part-Time Software Engineer

Leonardo UK

Jul. 2026 – Present

Newcastle upon Tyne, UK

- Selected as one of three undergraduate software engineers to continue at Leonardo part-time while completing my degree
- Develop safety-critical, real-time software remotely within an Agile engineering team alongside full-time university study
- Maintain and monitor Jenkins CI pipelines, supporting reliable automated build and test execution across ongoing development

Industrial Placement Software Engineer

Leonardo UK

Jun. 2025 – Jul. 2026

Newcastle upon Tyne, UK

- Developed and maintained safety-critical, real-time embedded software within an Agile engineering team, following DO-178C development practices
- Led the design and implementation of a C++ synthetic testing framework, reducing integration time by over 50% and enabling significantly more accurate project time estimates
- Led requirements and design workshops with 10+ engineers, including principal and lead engineers, to gather stakeholder needs and refine the software design
- Built a custom test application adopted by 30+ engineers across teams, enabling hardware-in-the-loop testing of complex systems
- Created a tool to visualise ultra-high-volume, real-time IQ data, enabling analysis of end to end system behaviour

PROJECTS

Atlas | C++, Vulkan, CMake, Python

Current Project

- Developing a user-space heterogeneous task scheduler supporting CPU workloads and preemptive-style GPU scheduling through cooperative execution slices using Vulkan.
- Designed and implemented multiple scheduling policies, including FIFO, priority-based and round-robin, with benchmarking to evaluate latency, throughput and fairness
- Built a Vulkan compute backend enabling fine-grained control over GPU work submission, profiling execution using timeline visualisation and performance metrics
- Built an automated benchmarking framework to evaluate scheduling overhead and performance across different workloads, including CPU-bound, GPU-bound and mixed workloads

Blocky | C++, OpenGL, GLSL, Python, CMake

Oct. 2022 – May 2023

- Developed a voxel-based 3D game engine in C++ using OpenGL and GLSL
- Achieved a stable 60 FPS while rendering over 100,000 voxels with real-time lighting
- Created a custom linear algebra library to support vector and matrix mathematics required for 3D graphics

TECHNICAL SKILLS

Languages: C++, C, Python, Haskell, Java, Matlab, VHDL, ARM, Rust, Shell Scripting, CMake

Libraries: OpenGL, Vulkan, Flask

Tools: Git, Docker, VS Code, Visual Studio, IntelliJ, Jenkins, MISRA, PolySpace

Training/Certificates: DO-178C Certified, C++ Training

Hobbies: Climbing, Bouldering, Leadership, Drones